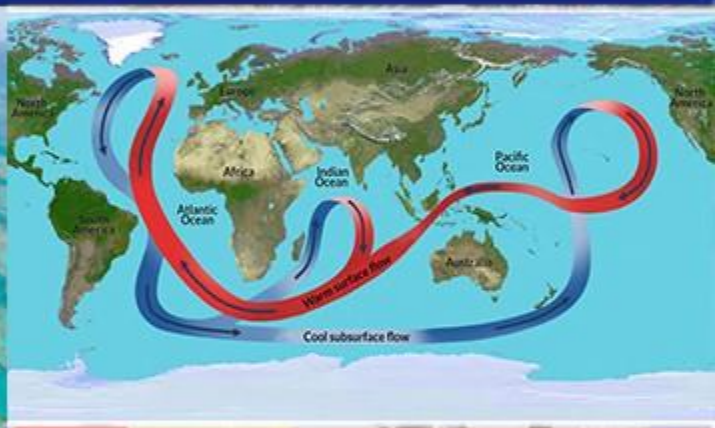


# GEOGRAPHY

# PHYSICAL GEOGRAPHY



# OCEANOGRAPHY

MODULE - 3

SALINITY



# OCEANOGRAPHY

## SALINITY OF OCEAN WATERS

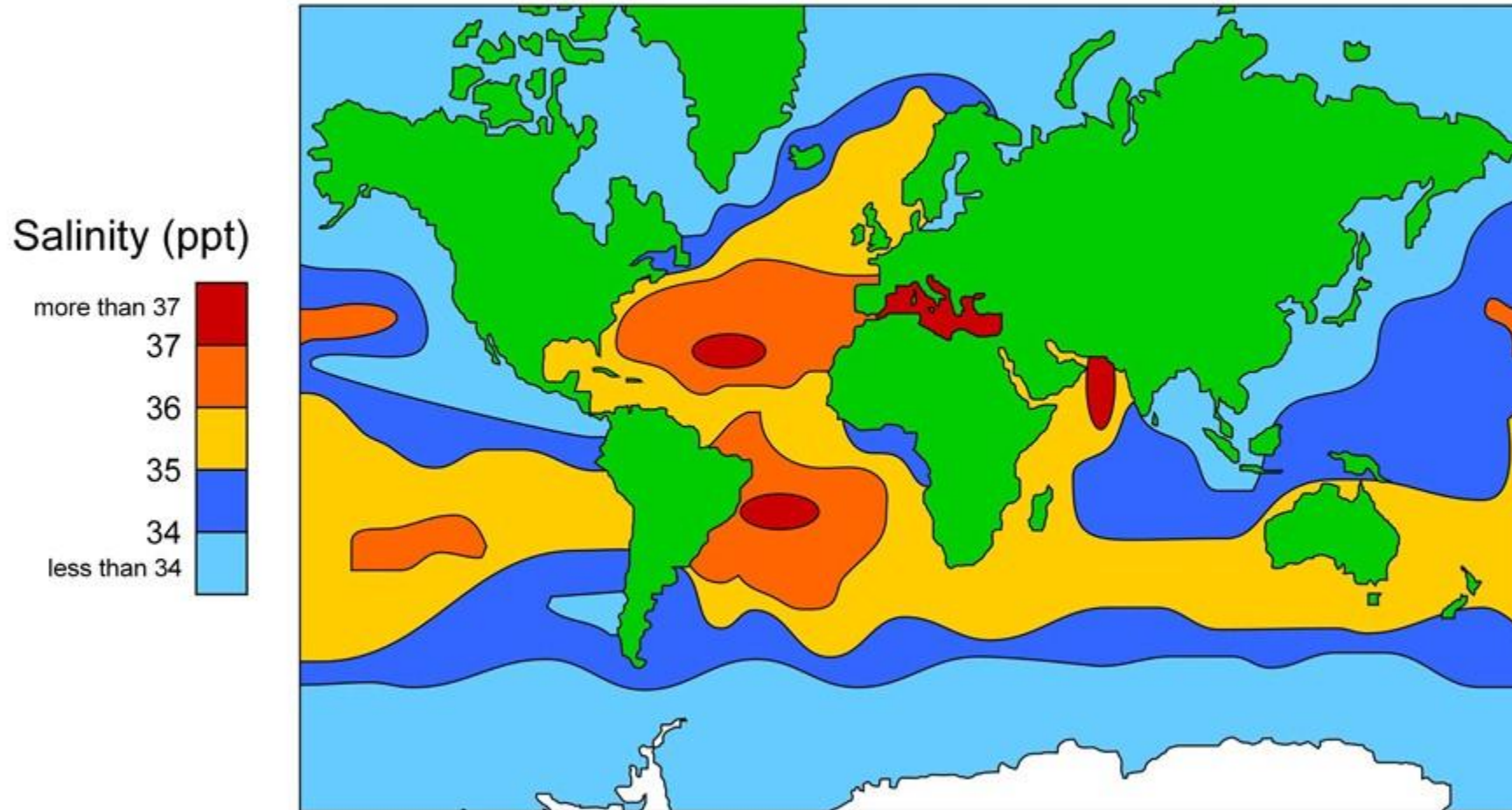
### SHORT NOTES ON SALINITY



- 1 All waters in nature, whether rain water or ocean water, contain **dissolved mineral salts**.
- 2 Salinity is the term used to **define** the total content of dissolved salts in sea water.
- 3 It is calculated as the **amount of salt (in gm)** dissolved in 1,000 gm (1 kg) of seawater.
- 4 It is usually expressed **as parts per thousand (‰) or ppt**. Salinity is an important property of sea water.
- 5 Salinity of 24.7 ‰ has been considered as the upper limit to demarcate 'brackish water'.

# OCEANOGRAPHY

## SALINITY OF OCEAN WATERS



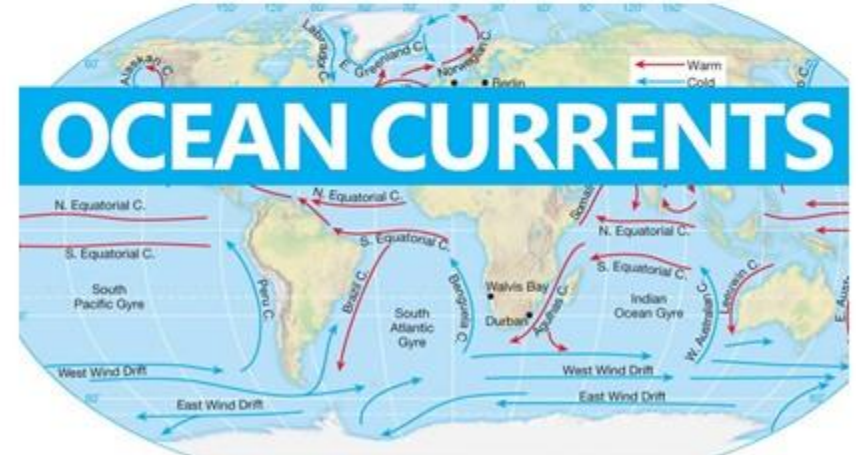
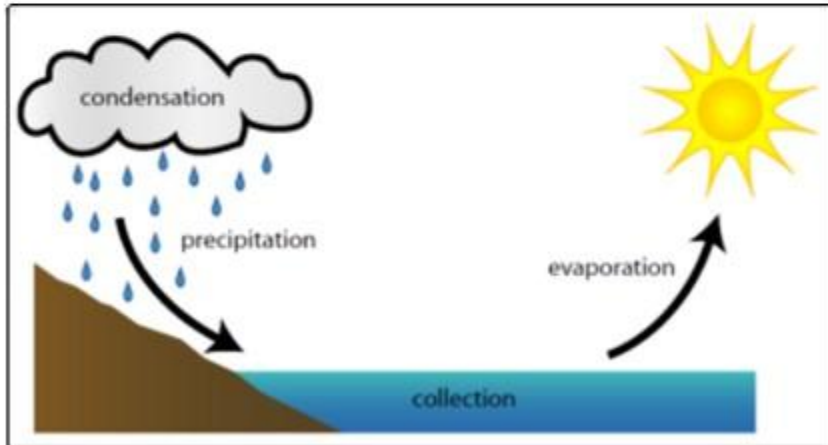
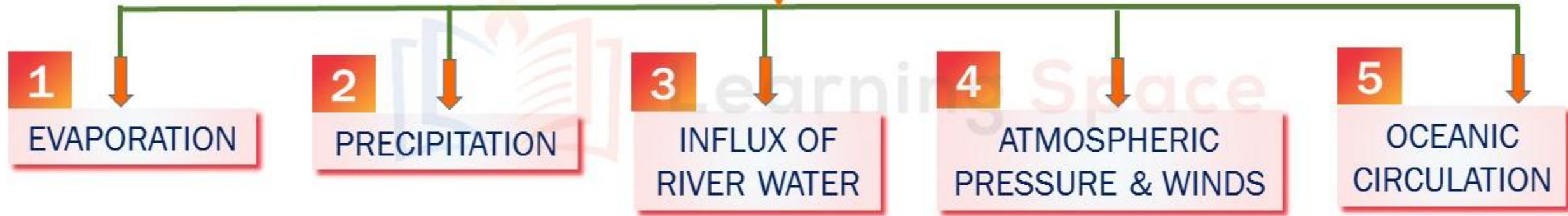
*On an average, seawater in the world's oceans has a salinity of approximately 3.5%, or 35 parts per thousand.*



# OCEANOGRAPHY

## SALINITY OF OCEAN WATERS

### FACTORS AFFECTING OCEAN SALINITY



# OCEANOGRAPHY

## SALINITY OF OCEAN WATERS

**1** The salinity of water in the surface layer of oceans depend **mainly on evaporation and precipitation.**

**2** Surface salinity is greatly influenced in **coastal regions** by the fresh water flow from rivers.

**3** Surface salinity in **polar** regions is influenced by the **processes of freezing and thawing of ice, inflow of glacier etc.**



**4** **Wind**, also influences salinity of an area by transferring water to other areas.

**5** The **ocean currents** too contribute to the salinity variations.

**6** Salinity, **temperature and density** of water are interrelated. Hence, any change in the temperature or density influences the salinity of water in an area.



# OCEANOGRAPHY

## SALINITY OF OCEAN WATERS

- 7** Highest salinity in water bodies Lake Van in Turkey (330 ‰), Dead Sea (238 ‰), Great Salt Lake (220 ‰).



DEAD SEA



LAKE VAN



GREAT SALT LAKE



# OCEANOGRAPHY

## SALINITY OF OCEAN WATERS

### SALINITY VARIATIONS IN THEORY

#### IN TROPICS

- 1 Thus salinity is higher where the addition of fresh water through rainfall, rivers and ice melt is lesser and the rate of evaporation is high.



- 2 Conversely the salinity is lower in areas receiving a lot of fresh water through rainfall, ice melt or rivers and where the loss of water through evaporation is lower.
- 3 In the open oceans, the salinity is highest in the tropical areas due to high rate of evaporation here.
- 4 The amount of rainfall in these regions is also lower and since large parts of the continents in these latitudes are deserts, not many rivers flow into the oceans and seas in these latitudes.



# OCEANOGRAPHY

## SALINITY OF OCEAN WATERS

### SALINITY VARIATIONS IN THEORY

#### IN TROPICS



Rainfall and inflow of rivers  
decreases salinity

- 5 Hence the loss of water through evaporation exceeds the addition of fresh water by means of rainfall or rivers etc.
- 6 From tropical areas salinity declines both towards the polar areas and the equatorial regions.
- 7 However, the decline is steeper towards the poles and in the equatorial regions it is only marginally lower than in the tropical latitudes.



# OCEANOGRAPHY

## SALINITY OF OCEAN WATERS

### SALINITY VARIATIONS IN THEORY

#### IN POLAR REGIONS



Ice melt in polar regions

- 8 The lower salinity in the polar areas is on account of two factors,
  - firstly, the **rate of evaporation** here is very low and
  - secondly, **a large amount of water** is added through ice melt.
- 9 Addition of fresh water into the oceans in these latitudes thus far exceeds the amount lost through evaporation.



# OCEANOGRAPHY

## SALINITY OF OCEAN WATERS

### SALINITY VARIATIONS IN THEORY

#### IN EQUATOR REGION



Cloud cover over oceans

**10** The chief cause of lower salinity in the equatorial regions is a large amount of rainfall here and a number of large rivers flowing into the seas.

**11** Also, the cloudy weather of these regions restricts the rate of water loss through evaporation.

**12** Thus here also the situation is similar to that prevailing in the polar areas, i.e. the addition of fresh water outstrips the loss due to evaporation.

## HORIZONTAL DISTRIBUTION OF SALINITY



- 1 The salinity for normal open ocean ranges between 33‰ and 37‰.
- 2 In the land locked Red Sea, it is as high as 41‰, while in the estuaries and the Arctic, the salinity fluctuates from 0 - 35‰, seasonally.
- 3 In hot and dry regions, where evaporation is high, the salinity sometimes reaches to 70‰.
- 4 Salinity gradually decreases from the tropics as we move northwards.



## HORIZONTAL DISTRIBUTION OF SALINITY



### SOME EXAMPLES OF SALINITY IN SEAS

- 5 The North Sea, in spite of its location in higher latitudes, records higher salinity due to more saline water brought by the warm North Atlantic Drift.
- 6 Baltic Sea records low salinity due to influx of river waters in large quantity.
- 7 The Mediterranean Sea records higher salinity due to high evaporation.



## HORIZONTAL DISTRIBUTION OF SALINITY



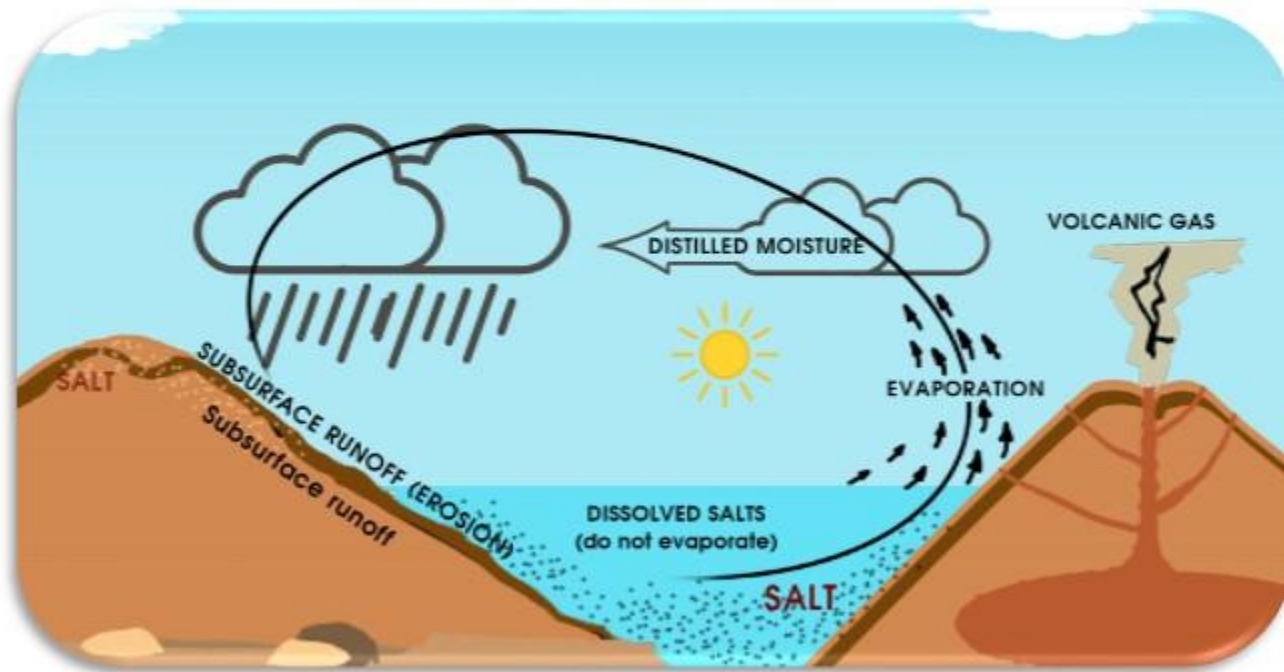
### SOME EXAMPLES OF SALINITY IN SEAS

- 8 Salinity is, however, very low in Black Sea due to enormous fresh water influx by rivers.
- 9 The average salinity of the Indian Ocean is 35‰.
- 10 The low salinity trend is observed in the Bay of Bengal due to influx of river water.
- 11 On the contrary, the Arabian Sea shows higher salinity due to high evaporation and low influx of fresh water.



## VERTICAL DISTRIBUTION OF SALINITY

- 1 Salinity changes with depth, but the way it changes depends upon the location of the sea.



- 2 Salinity at the surface increases by the loss of water to ice or evaporation, or decreased by the input of fresh waters, such as from the rivers.
- 3 Salinity at depth is more or less fixed, because there is no way that water is 'lost', or the salt is 'added.'
- 4 There is a marked difference in the salinity between the surface zones and the deep zones of the oceans.



## VERTICAL DISTRIBUTION OF SALINITY



- 5 The lower salinity water rests above the higher salinity dense water.

- 6 Salinity, generally, increases with depth and there is a distinct zone called the **halocline**, where salinity increases sharply.
- 7 Other factors being constant, increasing salinity of seawater causes its density to increase.
- 8 High salinity seawater, generally, sinks below the lower salinity water. This leads to **stratification by salinity**.

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